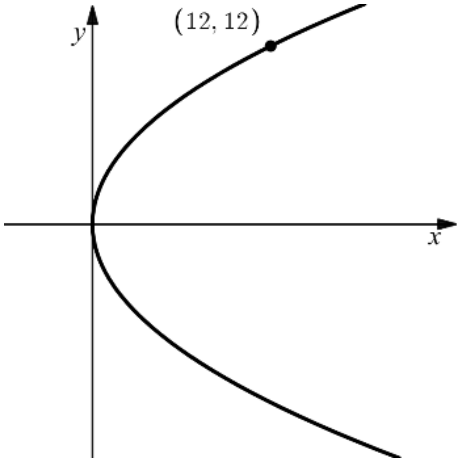
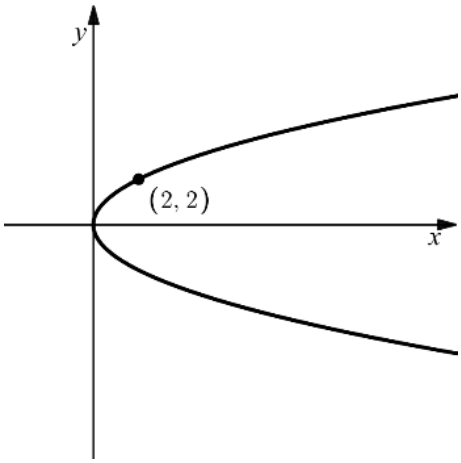
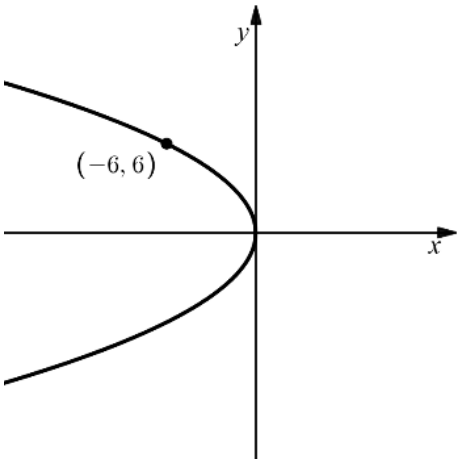
1) a)



1) b)



1) c)



2) a)

$\left(\sqrt{12}\right)^2 = 3a$

$\therefore a = 4$

2) b)

$\left(-\sqrt{2}\right)^2 = -a$

$\therefore a = -2$

2) c)

$6^2 = 6\sqrt{6}x$

$\therefore a = \sqrt{6}$

3) a)

intersection of $x = \frac{1}{4}y^2$ with $y = x$:

$y = \frac{1}{4}y^2$

$\Rightarrow y\left(1 - \frac{1}{4}y\right) = 0$

$\therefore y = 0$ or $y = 4$

$y^2 = 4x$ and $P(4, 4)$

3) b)

intersection of $x = 3y^2$ with $y = x$:

$y = 3y^2$

$\Rightarrow y(1 - 3y) = 0$

$\therefore y = 0$ or $y = \frac{1}{3}$

$y^2 = 4x$ and $P(4, 4)$

$y^2 = \frac{1}{3}x$ and $P\left(\frac{1}{3}, \frac{1}{3}\right)$

3) c)

intersection of $x = ay^2$ with $y = x$:

$y = ay^2$

$\Rightarrow y(1 - ay) = 0$

$\therefore y = 0$ or $y = \frac{1}{a}$

$y^2 = 4x$ and $P(4, 4)$

$y^2 = \frac{1}{a}x$ and $P\left(\frac{1}{a}, \frac{1}{a}\right)$

4) a)

$(2, 4)$

4) b)

$(4, 2)$

4) c)

$(5, -1)$

4) d)

$(-1, -5)$

5) a)

$a = 3$ and $b = -4$

5) b)

$(y - 3)^2 = x + 4$

When $y = 0$, $x = 5$

When $x = 0$, $y = 3 \pm 2$ so $y = 1$ or $y = 5$

$P(0,5)$, $Q(0, 1)$ and $R(5, 0)$